MD. ASHIQUR RAHMAN KHAN

Data Analyst | Public Health and Healthcare Analytics | B.Sc. in Statistics

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LinkedIn | GitHub | Portfolio

Summary

Statistics graduate from MBSTU working in public health and healthcare analytics. Built end-to-end projects in environmental health, dengue forecasting, US mortality and hospital readmission, using spatial analysis, time series forecasting, survival analysis, causal inference and machine learning. Works in R, Python, SQL and Power BI. Seeking a Data Analyst role in public health or healthcare, with a long-term goal of graduate study in Data Science and Machine Learning.

Projects

Fine Particle Pollution Exposure in Bangladesh (1998-2024)
2026 | *GitHub | Python, DuckDB SQL, R, XGBoost, SHAP, Power BI*

- Processed 27 years of ACAG satellite PM2.5 rasters with WorldPop to build population-weighted exposure for 8 divisions, 64 districts and 507 upazilas
- Found the average person breathes 67.6 µg/m^3 (13.5 times the WHO guideline), with 99.4% above the national standard in 2024; Mann-Kendall and Sen's slope showed the rise of +2.15 µg/m^3 a year stopped after 2010 (+0.06, p = 0.83)
- Measured strong spatial clustering (Moran's I 0.95, LISA, Gi*) and fitted SAR, SEM and SDM; the spatial error model cut residual Moran's I from 0.84 to 0.07
- Showed random cross-validation overstated XGBoost (R^2 0.94 vs 0.66 under spatial CV), estimated 6.1 years of life expectancy lost with the AQLI method, and built a 7-page Power BI dashboard with a 40-page report

Forecasting Dengue Hospital Admissions in Bangladesh
2026 | *GitHub | R, Python, DuckDB SQL, XGBoost, Power BI*

- Built weekly 1-4 week ahead forecasts for 8 divisions from 592,614 admissions (2023-2026), extracting counts from the DGHS dashboard's chart code
- Repaired about 105,000 missing 2023 admissions for Dhaka and Barishal from 176 daily DGHS press releases, with every rule logged
- Compared XGBoost, random forest, ARIMA with Fourier terms, ETS and negative binomial models on 2,912 rolling forecasts; only XGBoost beat the last-value forecast in both years (Diebold-Mariano p = 0.028 and 0.025)
- Showed most of the weather signal was seasonality (DLNM, correlation fell from 0.5-0.8 to 0.12 without the yearly cycle), and built an 8-page Power BI dashboard

Cardiometabolic Risk and Mortality in US Adults (NHANES)
2026 | *GitHub | Python, DuckDB SQL, R, Power BI*

- Followed 50,819 US adults (8,279 deaths) from NHANES linked to National Death Index mortality with survey-weighted Cox models over 20 imputations: diabetes HR 1.44, CKD HR 1.59, smoking HR 2.27
- Showed the crude prediabetes excess was age (adjusted HR 0.99), and that 1 minus Kaplan-Meier overstated cause-specific risk by up to 28.5%
- Used inverse probability weighting, built a 10-year risk model (C-index 0.872), an 8-page Power BI dashboard and a 50-page report

Diabetes 30-Day Hospital Readmission Analysis
2026 | *GitHub | Python, DuckDB SQL, XGBoost, SHAP, Power BI*

- Analyzed 69,970 diabetic patients from 130 US hospitals with SQL, logistic regression and propensity score matching: HbA1c testing had no overall effect (matched OR 0.99) but a significant interaction with diagnosis (p = 0.0067)
- Calibrated XGBoost model with SHAP (ROC-AUC 0.659): the top 20% by risk hold 38% of readmissions; 8-page Power BI dashboard, 158 DAX measures

Experience

Undergraduate Researcher - Department of Statistics
2024 - 2026 | *Mawlana Bhashani Science and Technology University (MBSTU)*

- Compared SARIMA with Random Forest, XGBoost, SVR, LSTM and a hybrid on 43 years of monthly NASA POWER wind speed for 64 districts (GitHub)
- Best holdout R-squared 0.83 (RMSE 0.236 m/s); Diebold-Mariano tests and rolling-origin cross-validation found no model significantly better than SARIMA
- Found and fixed a data leakage error in the rolling features, and wrote the results up in LaTeX with reproducible R and Python workflows

Languages

Bengali | English | Hindi

Education

B.Sc. in Statistics
Mawlana Bhashani Science and Technology University (MBSTU)
2021 - 2026, Tangail, Bangladesh
CGPA: 3.24 / 4.00

Skills

Programming Languages
R | Python | SQL | DAX | LaTeX

Public Health and Biostatistics
Epidemiology | Biostatistics | Survival Analysis | Cox Regression | Kaplan-Meier | Competing Risks | Survey-Weighted Analysis | Multiple Imputation | Causal Inference | Propensity Score Matching | Logistic Regression | Risk Prediction | Spatial Statistics | Spatial Regression

Statistics and Machine Learning
Hypothesis Testing | ANOVA | Chi-Square | Linear Regression | Bayesian Inference | Machine Learning | SARIMA | Time Series Forecasting | Predictive Modeling | Raster Processing

Libraries and Frameworks
pandas | NumPy | scikit-learn | XGBoost | SHAP | statsmodels | scipy | matplotlib | tidyverse | ggplot2 | survey | survival | mice | forecast | rasterio | spdep

Tools and Platforms
Power BI | Excel | Tableau | DuckDB | MySQL | SPSS | RStudio | Jupyter | Google Colab | Git | GitHub

Core Competencies
Data Cleaning | EDA | ETL | Data Modeling | Data Visualization | Dashboards | Report Writing | Communication | Problem Solving

Certifications

Power BI + Claude: AI-Powered Modeling with the MCP Connector
LinkedIn Learning | *2026*

Python for Data Science and Machine Learning Essential Training (Parts 1 and 2)
LinkedIn Learning | *2026*

Bayesian Statistics: From Concept to Data Analysis
UC Santa Cruz, Coursera | *2025*

Regression Models
Johns Hopkins University, Coursera | *2025*

Basic Programming with Python
Bangladesh Computer Council (EDGE) | *2025*

Data Analytics and Power BI Career Path
Interactive Cares | *2025*

Generative AI with Large Language Models
DeepLearning.AI and AWS, Coursera | *2025*

Generative AI for Everyone
DeepLearning.AI, Coursera | *2025*

Introduction to Data Analytics
IBM, Coursera | *2024*